

Probability Theory

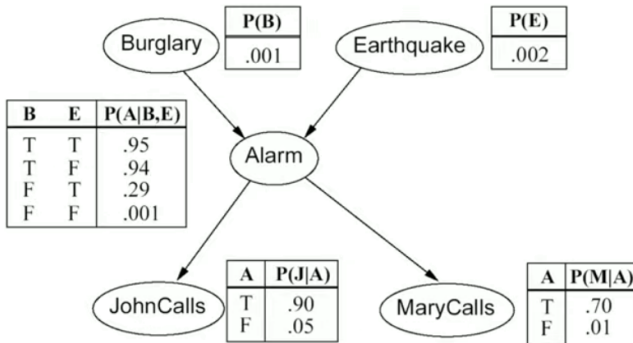
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- ① You have a new burglar alarm installed at home.
- ② It is fairly reliable at detecting burglaries, but it also sometimes responds to minor earthquakes.
- ③ You have two neighbors, John and Merry, who promised to call you at work whenever they hear the alarm.
- ④ John always calls when he hears the alarm, but sometimes he confuses the telephone ringing with the alarm and calls anyway.
- ⑤ Merry enjoys loud music and sometimes fails to hear the alarm.

Given the evidence of who has or has not called, we would like to estimate the situation.



Bayesian Network (BN)

A Bayesian Network (BN) is a **probabilistic graphical** model that represents a set of variables and their conditional dependencies via a **acyclic graph**.

A Bayesian Network is a probability graph. It uses Bayes' rule to calculate new probabilities for its variables as evidence flows through the graph.

Bayesian Neural Networks

Bayesian neural networks are a type of neural network that uses Bayesian inference to make predictions.

In a traditional neural network, the weights and biases of the network are fixed, and the model is trained to minimize the difference between the predicted output and the true output.

Given the name of a person, we want to predict the sex of a person.

For example, we have a person say 'Firdous'.

Classifying 'Firdous' as female or male is equivalent to asking is it more probable that 'Firdous' is male or female.

Mathematically, which one is greater $P(\text{male} \mid \text{Firdous})$ or $P(\text{female} \mid \text{Firdous})$

Let's apply Bayes theorem

$$P(\text{male} \mid \text{Firdous}) = \frac{P(\text{Firdous} \mid \text{male})P(\text{male})}{P(\text{Firdous})}$$

Diagram illustrating the components of Bayes' theorem for the example:

- $P(\text{Firdous} \mid \text{male})$: Probability of being named Firdous given male
- $P(\text{male})$: Probability of being male
- $P(\text{Firdous})$: Probability of being named Firdous



Causes of uncertainty:

Following are some leading causes of uncertainty to occur in the real world.

- ① Information occurred from unreliable sources.
- ② Experimental Errors
- ③ Equipment fault
- ④ Temperature variation
- ⑤ Climate change.

The “probability” of an event A

how frequently A is about to occur if we make many trials.

The word **probability** is used to mean the chance that a particular event (or set of events) will occur expressed on a linear scale from 0 (impossibility) to 1 (certainty), also expressed as a percentage between 0 and 100%

AI

Artificial Intelligence is about developing predictive models from uncertain data where uncertainty means working with imperfect or incomplete information. There are three main sources of uncertainty in AI, they are:

- 1 noisy data,
- 2 incomplete coverage of the problem domain
- 3 imperfect model

Basic Terminology in Probability

- ➊ **Random Experiment:** A process or procedure that generates outcomes in an unpredictable manner.
- ➋ **Outcome:** A result of a random experiment.
- ➌ **Sample Space:** The set of all possible outcomes of a random experiment.
- ➍ **Event:** A subset of the sample space.
- ➎ **Trial:** A single performance of a random experiment.
- ➏ **Probability:** The likelihood that an event will occur, expressed as a number between 0 and 1.
- ➐ **Equally Likely Outcomes:** Outcomes that have the same probability of occurring.
- ➑ **Mutually Exclusive Events:** Events that cannot occur simultaneously.
- ➒ **Independent Events:** Events that do not influence each other's occurrence.

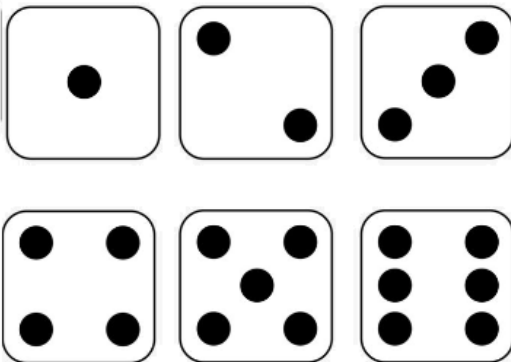
- ① **Random Experiment:** A process or procedure that generates outcomes in an unpredictable manner.
 - **Example:** Tossing a fair coin.
- ② **Outcome:** A result of a random experiment.
 - **Example:** When tossing a fair coin, the possible outcomes are Heads (H) or Tails (T).
- ③ **Sample Space:** The set of all possible outcomes of a random experiment.
 - **Example:** For a coin toss, the sample space is $\{H, T\}$.
- ④ **Event:** A subset of the sample space.
 - **Example:** The event of getting Heads (H) when tossing a fair coin is represented as $\{H\}$.
- ⑤ **Trial:** A single performance of a random experiment.
 - **Example:** Conducting one toss of a fair coin.
- ⑥ **Experiment:** A procedure that consists of one or more trials.
 - **Example:** Conducting five tosses of a fair coin.

- ① **Probability:** The likelihood that an event will occur, expressed as a number between 0 and 1.
 - **Example:** The probability of getting Heads (H) when tossing a fair coin is 0.5.
- ② **Equally Likely Outcomes:** Outcomes that have the same probability of occurring.
 - **Example:** In rolling a fair six-sided die, each number has an equal probability of $\frac{1}{6}$ of occurring.
- ③ **Mutually Exclusive Events:** Events that cannot occur simultaneously.
 - **Example:** When rolling a single die, the events of getting an even number (2, 4, 6) and getting an odd number (1, 3, 5) are mutually exclusive.
- ④ **Independent Events:** Events that do not influence each other's occurrence.
 - **Example:** Tossing a coin and rolling a die are independent events because the outcome of one does not affect the outcome of the other.

Some sample spaces for various probability experiments are shown here

Experiment	Sample space
Toss one coin	Head, tail
Roll a die	1, 2, 3, 4, 5, 6
Answer a true/false question	True, false
Toss two coins	Head-head, tail-tail, head-tail, tail-head

Ordinary Deck of Cards



Ordinary Deck of Cards



HEART



DIAMONDS



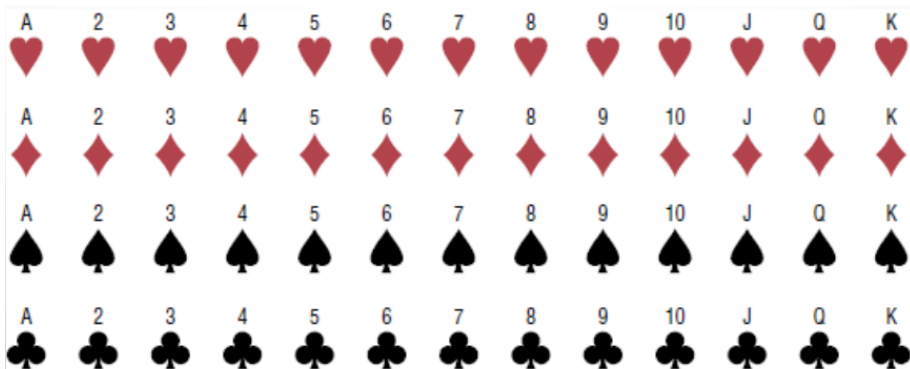
CLUB



SPADE

Ordinary Deck of Cards

Since there are 4 suits (hearts, diamonds, spades, and clubs) and 13 cards for each suit (ace through king), there are 52 outcomes in the sample space.



If the experiment consists of tossing two dice, then the sample space consists of the 36 points

$$S = \{(i, j) : i, j = 1, 2, 3, 4, 5, 6\}$$

where the outcome (i, j) is said to occur if i appears on the leftmost die and j on the other die.

If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consists of all nonnegative real numbers; that is,

$$S = \{x : 0 \leq x < \infty\}$$

Any subset E of the sample space is known as an event. In other words, an event is a set consisting of possible outcomes of the experiment

- 1 If $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice equals 7.
- 2 If $E = \{x : 0 \leq x \leq 5\}$, then E is the event that the transistor does not last longer than 5 hours

Classical probability

uses **sample spaces** to determine the numerical probability that an event will happen.

- You do not actually have to perform the experiment to determine that probability
- *Classical probability assumes that all outcomes in the sample space are equally likely to occur*
- Classical probability was the first type of probability studied formally by mathematicians in the 17th and 18th centuries.

Classical probability

Tossing a coin

If we flip a coin, then heads H and tails T will appear about equally often, we say that H and T are “equally likely.”

Tossing a Fair Die

For a regularly shaped die of homogeneous material (“fair die”) each of the six outcomes 1, 2, 3, 4, 5, 6 will be equally likely.

Selecting Card from Ordinary deck

When a card is selected from an ordinary deck of 52 cards, you assume that the deck has been shuffled, and each card has the same probability of being selected. In this case, it is $1/52$.

Formula for Classical Probability

The probability of any event E is

$$\frac{\text{Number of outcomes in } E}{\text{Total number of outcomes in the sample space}}$$

This probability is denoted by

$$P(E) = \frac{n(E)}{n(S)}$$

This probability is called classical probability, and it uses the sample space S .

Empirical or relative frequency probability

Empirical probability The difference between classical and empirical probability is that classical probability assumes that certain outcomes are equally likely (such as the outcomes when a die is rolled), while empirical probability relies on actual experience to determine the likelihood of outcomes. In empirical probability, one might actually roll a given die 6000 times, observe the various frequencies, and use these frequencies to determine the probability of an outcome.

Formula for Empirical probability Given a frequency distribution, the probability of an event being in a given class is

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

This probability is called empirical probability and is based on observation.

Classical vs Empirical Probability

Aspect	Classical Probability	Empirical Probability
Basis	Theoretical assumptions	Empirical observations or data
Application	Equally likely outcomes	Real-life situations or experiments
Sample Space	Finite, well-defined	Not necessarily finite or well-defined
Assumptions	All outcomes equally likely	No assumptions about outcomes
Calculation	$\frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$	$\frac{\text{Frequency of event E}}{\text{Total number of trials}}$
Accuracy	Accurate under assumptions	Depends on quality and quantity of data
Example	Rolling a fair six-sided die	Flipping a coin multiple times and recording outcomes

Example

A card is drawn from an ordinary deck. Find these probabilities.

- (a) Of getting a jack
- (b) Of getting the 6 of clubs (i.e., a 6 and a club)
- (c) Of getting a 3 or a diamond
- (d) Of getting a 3 or a 6

(a) $P(\text{jack}) = \frac{4}{52}$

(b) $P(\text{6 of clubs}) = \frac{1}{52}$

(c) $P(\text{3 or diamond}) = \frac{16}{52}$

(d) $P(\text{3 or a 6}) = \frac{8}{52}$

Example

In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood, and 2 had type AB blood. Set up a frequency distribution and find the following probabilities.

- (i) A person has type O blood.
- (ii) A person has type A or type B blood.
- (iii) A person has neither type A nor type O blood.
- (iv) A person does not have type AB blood

Type	Frequency
A	22
B	5
AB	2
O	21
	Total 50

$$(i)=21/50, (ii)=27/50, (iii)=7/50, (iv)=1-\frac{2}{50}.$$

Basic Rules for Probability

Four Basic Rules for Probability

The probability of any event E is a number (either a fraction or decimal) between and including 0 and 1. This is denoted by

$$0 \leq P(E) \leq 1.$$

Rule 1 states that probabilities cannot be negative or greater than 1.

If an event E cannot occur (i.e., the event contains no members in the sample space), its probability is 0.

When a single die is rolled, find the probability of getting a 9.

If an event E is certain, then the probability of E is 1.

When a single die is rolled, what is the probability of getting a number less than 7?

The sum of the probabilities of all the outcomes in the sample space is 1.

$$\begin{aligned} 0 \leq P(E) \leq 1, \quad E \subseteq S \\ P(S) = 1, \quad \text{For entire space } S \\ \sum P(E) = 1 \end{aligned} \tag{1}$$

For example, in the roll of a fair die, each outcome in the sample space has a probability of $\frac{1}{6}$. Hence, the sum of the probabilities of the outcomes is as shown.

Outcome	1		2		3		4		5		6
Probability	$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$
Sum	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6} = \frac{6}{6} = 1$

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Outcome	1		2		3		4		5		6
Probability	$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$
Sum	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6}$	+	$\frac{1}{6} = \frac{6}{6} = 1$

$$\begin{aligned}
 0 &\leq P(E) \leq 1, \quad E \subseteq S \\
 P(S) &= 1, \quad \text{For entire space } S \\
 \sum P(E) &= 1
 \end{aligned}$$

Three horses A , B and C are in a race; A is twice as likely to win as B and B is twice as likely to win as C . What are their respective probabilities of winning?

Complementary Events

Complementary Events

Another important concept in probability theory is that of complementary events. When a die is rolled, for instance, the sample space consists of the outcomes 1, 2, 3, 4, 5, and 6. The event E of getting odd numbers consists of the outcomes 1, 3, and 5. The event of not getting an odd number is called the complement of event E , and it consists of the outcomes 2, 4, and 6.

Complement of an event E

The *complement of an event E* is the set of outcomes in the sample space that are not included in the outcomes of event E . The complement of E is denoted by $E^c = S \setminus E$.

Rule for Complementary Events

$$P(E^c) = 1 - P(E) \quad E \subseteq S. \quad (2)$$

If the probability of an event or the probability of its complement is known, then the other can be found by subtracting the probability from 1.

Let a coin and a die be tossed; let the sample space S consist of twelve elements

Let's denote the outcomes of tossing a coin and a die as follows:

- For the coin toss: H (Heads), T (Tails)
- For the die toss: 1, 2, 3, 4, 5, 6

The sample space is the set of all possible outcomes:

$$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), \\ (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$$

(i) Express explicitly the following events

$$A = \{\text{Heads and even numbers appear}\}$$

$$B = \{\text{A prime number appears}\}$$

$$C = \{\text{Tails and odd numbers appear}\}$$

(ii) Express explicitly that events

(a) A or B occurs

(b) B and C occurs

(c) Only B occurs

(iii) Which of the events A , B and C are mutually exclusive?

Five coins are tossed simultaneously. Find the probability of the event E : At least one head turns up. Assume that the coins are fair.

1	H	H	H	H	H
2	H	H	H	H	T
3	H	H	H	T	H
4	H	H	H	T	T
5	H	H	T	H	H
6	H	H	T	H	T
7	H	H	T	T	H
8	H	H	T	T	T
9	H	T	H	H	H
10	H	T	H	H	T
11	H	T	H	T	H
12	H	T	H	T	T
13	H	T	T	H	H
14	H	T	T	H	T
15	H	T	T	T	H
16	H	T	T	T	T
17	T	H	H	H	H
18	T	H	H	H	T
19	T	H	H	T	H
20	T	H	H	T	T
21	T	H	T	H	H
22	T	H	T	H	T
23	T	H	T	T	H
24	T	H	T	T	T
25	T	T	H	H	H
26	T	T	H	H	T
27	T	T	H	T	H
28	T	T	H	T	T
29	T	T	T	H	H
30	T	T	T	H	T
31	T	T	T	T	H
32	T	T	T	T	T

¹that is TTTTT

Five coins are tossed simultaneously. Find the probability of the event E : At least one head turns up. Assume that the coins are fair.

1	H	H	H	H	H
2	H	H	H	H	T
3	H	H	H	T	H
4	H	H	H	T	T
5	H	H	T	H	H
6	H	H	T	H	T
7	H	H	T	T	H
8	H	H	T	T	T
9	H	T	H	H	H
10	H	T	H	H	T
11	H	T	H	T	H
12	H	T	H	T	T
13	H	T	T	H	H
14	H	T	T	H	T
15	H	T	T	T	H
16	H	T	T	T	T
17	T	H	H	H	H
18	T	H	H	H	T
19	T	H	H	T	H
20	T	H	H	T	T
21	T	H	T	H	H
22	T	H	T	H	T
23	T	H	T	T	H
24	T	H	T	T	T
25	T	T	H	H	H
26	T	T	H	H	T
27	T	T	H	T	H
28	T	T	H	T	T
29	T	T	T	H	H
30	T	T	T	H	T
31	T	T	T	T	H
32	T	T	T	T	T

Since each coin can turn up heads or tails, the sample space consists of 32 outcomes. Since the coins are fair, we may assign the same probability to each outcome. Then the event (No heads turn up) consists of only 1 outcome¹. Hence $P(E^c) = 1/32$. Thus $P(E) = 1 - \frac{1}{32} = \frac{31}{32}$.

¹that is TTTTT

Hospital records indicated that Dengue patients stayed in the hospital for the number of days shown in the distribution.

Number of days	Frequency
2	8
3	15
4	32
5	56
6	19
7	5
8	8
9	10
Total	153

Find these probabilities.

- (a) A patient stayed less than 8 days.
- (b) A patient stayed exactly 5 days.
- (c) A patient stayed less than 6 days.
- (d) A patient stayed at most 4 days.
- (e) A patient stayed at least 5 days.

Using Direct Calculation

To find the probability that a patient stayed less than 8 days ($P(\text{less than 8 days})$), we sum up the frequencies of patients who stayed for 2, 3, 4, 5, 6, and 7 days, and then divide by the total number of patients:

$$P(\text{less than 8 days}) = \frac{\text{Freq of patients staying for 2, 3, 4, 5, 6, or 7 days}}{\text{Total number of patients}}$$

From the given data: $P(\text{less than 8 days}) = \frac{135}{153} \approx 0.8824$

Using Complement

To find $P(\text{less than 8 days}) = P(E)$, we can use the complement $P(E) = 1 - P(E')$:

$$P(E') = \frac{\text{Frequency of patients staying for 8 or more days}}{\text{Total number of patients}}$$

From the given data: $P(E') = \frac{18}{153} \approx 0.1176$

Now, we can find $P(E)$:

$$P(E) = 1 - P(E') = 1 - 0.1176 = 0.8824$$

So, the probability that a patient stayed less than 8 days is approximately 0.8824 or 88.24%.

The Addition Rules for Probability

Mutually Exclusive Events

Two events are mutually exclusive events if they cannot occur at the same time (i.e., they have no outcomes in common). *Two events A and B are mutually exclusive events if $A \cap B = \emptyset$.*

Determine which events are mutually exclusive and which are not, when a single die is rolled.

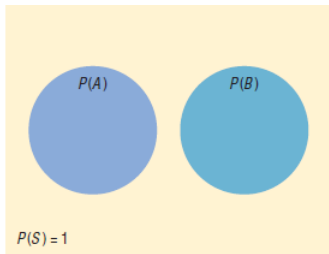
- a. Getting an odd number and getting an even number
- b. Getting a 3 and getting an odd number
- c. Getting an odd number and getting a number less than 4
- d. Getting a number greater than 4 and getting a number less than 4

Probability for Mutually Exclusive Events When two events A and B are mutually exclusive, the probability that A or B will occur is

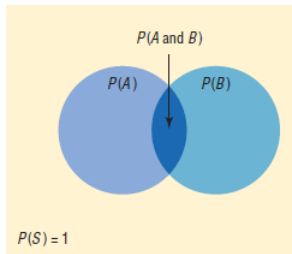
$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) \quad (3)$$

Probability for non-Mutually Exclusive Events When two events A and B are mutually exclusive, the probability that A or B will occur is

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (4)$$



(a) Mutually exclusive events
 $P(A \text{ or } B) = P(A) + P(B)$



(b) Nonmutually exclusive events
 $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

- 1 A day of the week is selected at random. Find the probability that it is a weekend day.

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**

$$P(\text{Saturday or Sunday}) = 2/7$$

- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either a king or a black card.

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**
 $P(\text{Saturday or Sunday}) = 2/7$
- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either a king or a black card. **Solution:**
 $P(\text{king or black card}) = P(\text{king}) + P(\text{black card}) - P(\text{black king})$

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**

$$P(\text{Saturday or Sunday}) = 2/7$$

- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either a king or a black card. **Solution:**

$$P(\text{king or black card}) = P(\text{king}) + P(\text{black card}) - P(\text{black king}) = 4/52 + 26/52 - 2/52 = 7/13.$$

For three events that are not mutually exclusive, $P(A \text{ or } B \text{ or } C) =$

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) \\ & - \{P(A \cap B) + P(B \cap C) + P(C \cap A)\} \\ & + P(A \cap B \cap C) \end{aligned}$$

A class contains 10 men and 20 women of which half the men and half the women have brown eyes. Find the probability p that a person is chosen at random is a man or has brown eyes.

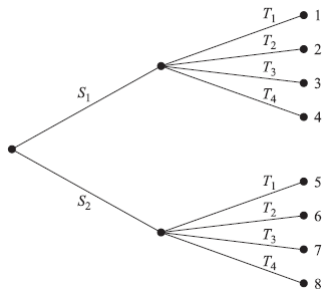
$$P\{\text{man or brown eye}\} = 2/3$$

Tree Diagram

A tree diagram is a device consisting of line segments emanating from a starting point and also from the outcome point. It is used to determine all possible outcomes of a probability experiment.

Example

Letting the shirts be represented by S_1 , S_2 and the ties by T_1, T_2, T_3, T_4 , the various ways of choosing a shirt and then a tie are indicated in the tree diagram of Fig



Dependent and Independent Events

Dependent Events and Their Probability

Definition

Events are dependent if the occurrence of one event affects the probability of the other.

Example

Consider drawing two cards from a standard deck without replacement.
Let:

- A : Drawing a King on the first draw
- B : Drawing a Queen on the second draw, given that a King was drawn on the first draw

We want to find $P(A \text{ and } B)$.

Probability of Dependent Events

If A and B are dependent events:

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

- Probability of drawing a King: $P(A) = \frac{4}{52}$
- Probability of drawing a Queen after drawing a King: $P(B|A) = \frac{4}{51}$
- Therefore, $P(A \text{ and } B) = \frac{4}{52} \times \frac{4}{51}$

Result

$$P(A \text{ and } B) = \frac{16}{2652} \approx 0.0060$$

The Multiplication Rules and Conditional Probability

The Multiplication Rules and Conditional Probability

Independent Events (Multiplication Rule 1)

When two events are independent, the probability of both occurring is

$$P(A \text{ and } B) = P(A \cap B) = P(A).P(B) \quad (5)$$

Dependent Events (Multiplication Rule 2)

When two events are dependent, the probability of both occurring is

$$P(A \text{ and } B) = P(A \cap B) = P(A)P(B/A) \quad P(B/A) = \frac{P(A \cap B)}{P(A)} \quad (6)$$

$P(B/A)$ denotes

the probability of B given that event A has occurred.

Here are some examples of dependent events:

Drawing a card from a deck, not replacing it, and then drawing a second card.

Selecting a ball from an urn, not replacing it, and then selecting a second ball.

Example 1: Rolling Two Dice

- Roll two six-sided dice.
- Event A : Rolling a 4 on the first die.
- Event B : Rolling a 6 on the second die.
- The events are independent.

$$P(A) = \frac{1}{6}, \quad P(B) = \frac{1}{6}$$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

- The probability of rolling a 4 on the first die and a 6 on the second die is $\frac{1}{36}$.

Example 2: Tossing a Coin and Rolling a Die

- Toss a coin and roll a six-sided die.
- Event A : Getting heads on the coin.
- Event B : Rolling a 5 on the die.
- The events are independent.

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{6}$$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

- The probability of getting heads on the coin and rolling a 5 on the die is $\frac{1}{12}$.

Example 3: Drawing Marbles Without Replacement

- A bag contains 5 red and 3 blue marbles.
- Draw two marbles without replacement.
- Event A : Drawing a red marble first.
- Event B : Drawing a red marble second.
- The events are dependent.

$$P(A) = \frac{5}{8}, \quad P(B|A) = \frac{4}{7}$$

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{5}{8} \cdot \frac{4}{7} = \frac{5}{14}$$

- The probability of drawing two red marbles is $\frac{5}{14}$.

Example 4: Picking Cards Without Replacement

- Draw two cards from a deck of 52 without replacement.
- Event A : Drawing a king on the first draw.
- Event B : Drawing a queen on the second draw.
- The events are dependent.

$$P(A) = \frac{1}{13}, \quad P(B|A) = \frac{4}{51}$$

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{1}{13} \cdot \frac{4}{51} = \frac{4}{663}$$

- The probability of drawing a king and then a queen is $\frac{4}{663}$.

For any three events A_1 , A_2 , A_3 , we have

$$P(A_1 \cap A_2 \cap A_3) = P(A_1).P(A_2/A_1).P(A_3/(A_1 \cap A_2))$$

In words, the probability that A_1 and A_2 and A_3 all occur is equal to the probability that A_1 occurs times the probability that A_2 occurs given that A_1 has occurred times the probability that A_3 occurs given that both A_1 and A_2 have occurred.

Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events.

- ① Getting 3 jacks
- ② Getting an ace a king, and a queen in order
- ③ Getting a club, a spade, and a heart in order
- ④ Getting 3 clubs

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Solution:

$$(a) P(3 \text{ jacks}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{1}{5525}$$

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$$(b) P(\text{ace and king and queen}) = \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50}$$

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$$(c) P(\text{club and spade and heart}) = \frac{13}{52} \cdot \frac{13}{51} \cdot \frac{13}{50}$$

$$(d) P(3 \text{ clubs}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50}$$

Two cards are drawn from a well-shuffled ordinary deck of 52 cards. Find the probability that they are both aces if the first card is (a) replaced, (b) not replaced.

- ① Find the probability that a single toss of a die will result in a number less than 4 if
 - (a) no other information is given
 - (b) it is given that the toss resulted in an odd number.

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No other information is given

Outcome : $\{1, 2, 3, 4, 5, 6\}$

Event : $\{1, 2, 3\}$

$$P(\text{Event}) = \frac{n(\text{Event})}{n(\text{outcome})} = \frac{3}{6}$$

It is given that the toss resulted in an odd number

Outcome : $\{1, 2, 3, 4, 5\}$

Event : $\{1, 3, 5\}$

P(Event)

Part (b): Given Result is Odd

Event $B = \{1, 3, 5\}$ (odd number)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$A \cap B = \{1, 3\}, \text{ so } P(A \cap B) = \frac{2}{6}$$

$$P(B) = \frac{3}{6}$$

$$P(A|B) = \frac{2/6}{3/6} = \frac{2}{3}$$

Example: Independent Events

Problem

Consider rolling two fair six-sided dice. Find the probability of rolling a 4 on the first die and a 6 on the second die.

- Let A be the event of rolling a 4 on the first die.
- Let B be the event of rolling a 6 on the second die.
- Since each die roll is independent:

$$P(A) = \frac{1}{6} \quad (\text{probability of rolling a 4})$$

$$P(B) = \frac{1}{6} \quad (\text{probability of rolling a 6})$$

- Therefore, $P(A \text{ and } B) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

Example: Dependent Events

Problem

A bag contains 4 red balls and 6 blue balls. Two balls are drawn without replacement. Find the probability of drawing a red ball followed by a blue ball.

- Let A be the event of drawing a red ball first.
- Let B be the event of drawing a blue ball second, given that a red ball was drawn first.
- $P(A) = \frac{4}{10}$ (probability of drawing a red ball initially)
- After drawing a red ball, there are 9 balls left, so $P(B|A) = \frac{6}{9}$.
- Therefore, $P(A \text{ and } B) = \frac{4}{10} \times \frac{6}{9} = \frac{4}{15}$.

Example: Dependent Events with Replacement

Problem

A jar contains 3 red balls and 5 blue balls. A ball is drawn from the jar, its color noted, and then it's replaced. Then another ball is drawn. Find the probability of drawing two blue balls.

- Let A be the event of drawing a blue ball on the first draw.
- Let B be the event of drawing a blue ball on the second draw, given that a blue ball was drawn on the first draw.
- $P(A) = \frac{5}{8}$ (probability of drawing a blue ball initially)
- Since the ball is replaced, $P(B|A) = \frac{5}{8}$ as well.
- Therefore, $P(A \text{ and } B) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$.

- ① A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss.

Let A_1 be the event “4, 5, or 6 on first toss,” and A_2 be the event “1, 2, 3, or 4 on second toss.” Then we are looking for $P(A_1 \cap A_2)$.

Method#1 Each of the 3 ways in which A_1 can occur can be associated with each of the 4 ways in which A_2 can occur to give $3 \cdot 4 = 12$ ways in which both A_1 and A_2 can occur out of total 36 ways. Then

$$P(A_1 \cap A_2) = \frac{12}{36}$$

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Method#2

$$P(A_1 \cap A_2) = P(A_1)P(A_2/A_1)$$

The fact that the result of the second toss is independent of the first so that $P(A_2/A_1) = P(A_2)$. Thus

$$P(A_1 \cap A_2) = P(A_1)P(A_2) =$$

Suppose that our sample space S is the population of adults in a small town who have completed the requirements for a college degree. We shall categorize them according to gender and employment status. The data are given in Table

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

One of these individuals is to be selected at random find the probability that the selected person is **employed** and male.

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Suppose M : a male is chosen and E : the chosen one is employed. Find

$$P(M/E)$$

A recent survey asked 100 people if they thought women in the armed forces should be permitted to participate in combat. The results of the

survey are shown.

Gender	Yes	No	Total
Male	32	18	50
Female	8	42	50
Total	40	60	100

Find these probabilities.

- The respondent answered yes, given that the respondent was a female.
- The respondent was a male, given that the respondent answered no.

Solution:

Let

M = respondent was a male
 F = respondent was a female

Y = respondent answered yes
 N = respondent answered no

Gender	Yes	No	Total
Male	32	18	50
Female	8	42	50
Total	40	60	100

The respondent answered yes, given that the respondent was a female

$$P(Y/F) = \frac{P(F \text{ and } Y)}{P(F)}$$

The probability $P(F \text{ and } Y)$ is the number of females who responded yes, divided by the total number of respondents:

$$P(Y/F) = \frac{P(F \text{ and } Y)}{P(F)} = \frac{\frac{n(F \cap Y)}{n(S)}}{\frac{n(F)}{n(S)}} = \frac{8/100}{50/100} = \frac{4}{25}$$

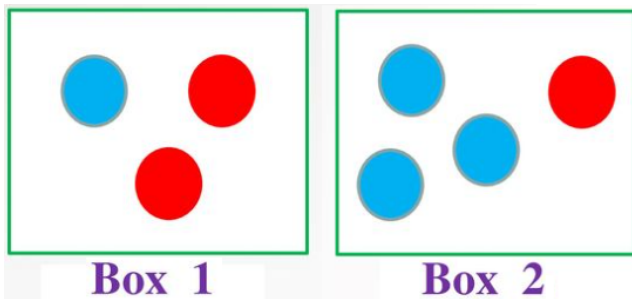
$$P(M/N) = \frac{P(M \text{ and } N)}{P(N)} = \frac{18/100}{60/100} = \frac{3}{10}$$

A regional telephone company operates three identical relay stations at different locations. During a one-year period, the number of malfunctions reported by each station and the causes are shown below.

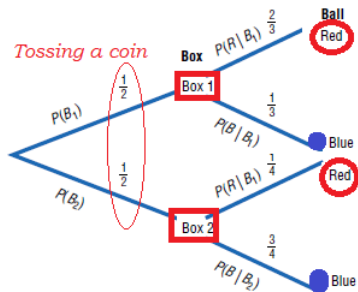
Station	A	B	C
Problems with electricity supplied	2	1	1
Malfunctioning electrical equipment	5	4	2
Computer malfunction	4	3	2
Caused by other human errors	7	7	5

Suppose that a malfunction was reported and it was found to be caused by other human errors. What is the probability that it came from station C?

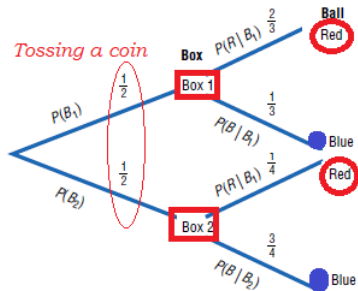
Box-1 contains 2 red balls and 1 blue ball. Box-2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, Box-1 is selected and a ball is drawn. If it falls tails up, Box-2 is selected and a ball is drawn. Find the probability of selecting a red ball.



Box-1 contains 2 red balls and 1 blue ball. Box-2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, Box-1 is selected and a ball is drawn. If it falls tails up, Box-2 is selected and a ball is drawn. Find the probability of selecting a red ball.



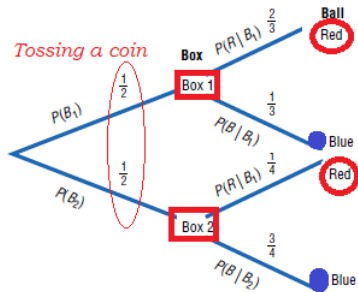
Box-1 contains 2 red balls and 1 blue ball. Box-2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, Box-1 is selected and a ball is drawn. If it falls tails up, Box-2 is selected and a ball is drawn. Find the probability of selecting a red ball.



How can we get red ball?

- 1 Selecting Box. " Box-1 or Box-2"
- 2 Find red ball.

Box-1 contains 2 red balls and 1 blue ball. Box-2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, Box-1 is selected and a ball is drawn. If it falls tails up, Box-2 is selected and a ball is drawn. Find the probability of selecting a red ball.



How can we get red ball?

- ① Selecting Box. " Box-1 or Box-2"
- ② Find red ball.

$$P(\text{red ball}) = P(\text{red ball from Box-1}) + P(\text{red ball from Box-2})$$

- Let B_1 be the event of selecting Box-1.
- Let B_2 be the event of selecting Box-2.
- Let R be the event of drawing a red ball.

$$P(R) = \underbrace{P(B_1)}_{\text{Heads up}} \cdot P(R|B_1) + \underbrace{P(B_2)}_{\text{Tails up}} \cdot P(R|B_2)$$

$$P(R|B_1) = \frac{2}{3}, \quad P(R|B_2) = \frac{1}{4}, \quad P(B_1) = \frac{1}{2}, \quad P(B_2) = \frac{1}{2}$$

Therefore,

$$P(R) = \frac{2}{3} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2} = \frac{11}{24}$$

Result

The probability of selecting a red ball is $\frac{11}{24}$.

Next determine the probabilities for each branch. Since a coin is being tossed for the box selection, each branch has a probability of $\frac{1}{2}$ that is, heads for box 1 or tails for box 2. The probabilities for the second branches are found by using the basic probability rule. For example, if box 1 is selected and there are 2 red balls and 1 blue ball, the probability of selecting a red ball is $\frac{2}{3}$ and the probability of selecting a blue ball is $\frac{1}{3}$. If box 2 is selected and it contains 3 blue balls and 1 red ball, then the probability of selecting a red ball is $\frac{1}{4}$ and the probability of selecting a blue ball is $\frac{3}{4}$. Next multiply the probability for each outcome, using the rule $P(\text{red and Box} - 1) = P(\text{Box} - 1) \cdot P(\text{red} / \text{Box} - 1)$. For example, the probability of selecting box 1 and selecting a red ball is

$$\frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6}.$$

The probability of selecting box 2 and selecting a red ball is obtained using the rule $P(\text{red and Box} = 2) = P(\text{Box} = 2) \cdot P(\text{red} / \text{Box} = 2)$.

$$\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

The probability of selecting box 2 and a blue ball is (Note that the sum of these probabilities is 1.)

Finally a red ball can be selected from either box 1 or box 2 so

$$P(\text{red}) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}.$$

In the a previous example, there were two boxes, each containing red balls and blue balls. A box was selected and a ball was drawn. The example asked for the probability that the ball selected was red.

$$P(\text{red ball}) = P(B1)P(R/B1) + P(B2)P(R/B2) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}$$

Now, a different question can be asked:

If the ball is red, what is the probability it came from box 1? In this case, the outcome is known, a red ball was selected, and you are asked to find the probability that it is a result of a previous event, that it came from box 1.

$$P(\text{Box1 given red ball}) = P(B1/R) = \frac{P(B1 \cap R)}{P(R)}$$

Bays Theorem

Consider the data according to gender and employment status as follows

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

- 1 Suppose that an additional information that 36 of those employed and 12 of those unemployed are members of the Army Club, is given.
- 2 We wish to find the probability of the event A that the individual selected is a member of the Army Club.

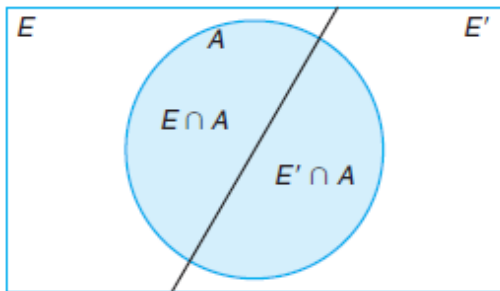
Let E denotes the set of employed persons the E^c represents unemployed

$E \cap A$ represents "Employed members of A "

$E^c \cap A$ represents "Unemployed members of A "

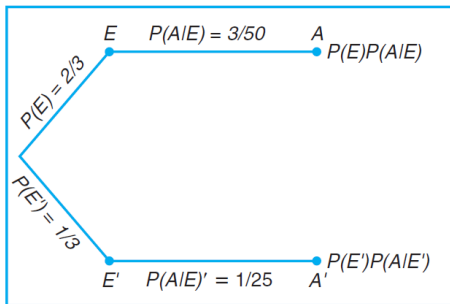
A is union of two mutually exclusive events $E \cap A$ and $E^c \cap A$

$$\begin{aligned} P(A) &= P[(E \cap A) \cup (E^c \cap A)] = P(E \cap A) + P(E^c \cap A) \\ &= P(E)P(A/E) + P(E^c)P(A/E^c) \end{aligned}$$



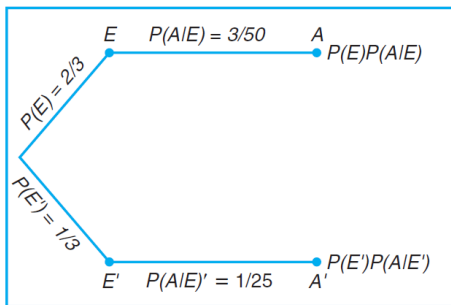
$$P(E) = \frac{600}{900} = \frac{2}{3}, \quad P(A|E) = \frac{36}{600} = \frac{3}{50},$$

$$P(U) = \frac{300}{900} = \frac{1}{3}, \quad P(A|U) = \frac{12}{300} = \frac{1}{25}.$$



$$P(E) = \frac{600}{900} = \frac{2}{3}, \quad P(A|E) = \frac{36}{600} = \frac{3}{50},$$

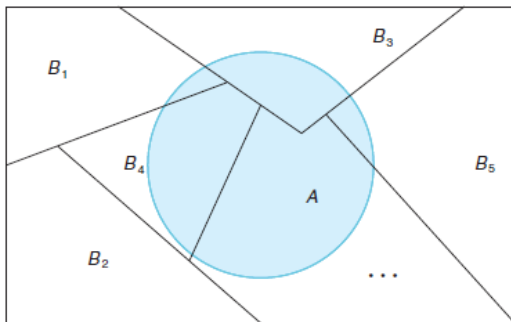
$$P(U) = \frac{300}{900} = \frac{1}{3}, \quad P(A|U) = \frac{12}{300} = \frac{1}{25}.$$



$$P(A) = \frac{2}{3} \times \frac{3}{50} + \frac{1}{3} \times \frac{1}{25} = \frac{4}{75}$$

If the events $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S such that $P(B_i) \neq 0$ for $i = 1, 2, \dots, k$, then for any event A of S ,

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i)P(A/B_i).$$



Bayes' Theorem or Rule

In a certain assembly plant, three machines, B_1 , B_2 , and B_3 , make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected.

- ① What is the probability that it is defective?
- ② If a product was chosen randomly and found to be defective, what is the probability that it was made by machine B_3 ?

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- ① $P(\text{the product is defective})$
 - ② $P(?)$ In this case, the outcome is known, the product is defective, and you are asked to find the probability that it came from B_3

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- 1 $P(D) = P(B_1)P(D/B_1) + P(B_2)P(D/B_2) + P(B_3)P(D/B_3)$
- 2 Find $P(B_3/D)$?

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- 2 Find $P(B_3/D)$?

$$P(B_3/D) = \frac{P(B_3 \cap D)}{P(D)} = \frac{P(B_3)P(D/B_3)}{P(B_1)P(D/B_1) + P(B_2)P(D/B_2) + P(B_3)P(D/B_3)}$$

Bayes' Theorem or Rule

Theorem

Suppose that $B_1, B_2, \dots, B_n$ are **mutually exclusive** events whose union is the sample space S , i.e., one of the events must occur. Then if D is any event, we have

$$P(B_k/D) = \frac{P(B_k)P(D/B_k)}{\sum_{j=1}^n P(B_j)P(D/B_j)}$$

In the a previous example, there were two boxes, each containing red balls and blue balls. A box was selected and a ball was drawn. The example asked for the probability that the ball selected was red.

$$P(\text{red ball}) = P(B1)P(R/B1) + P(B2)P(R/B2) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}$$

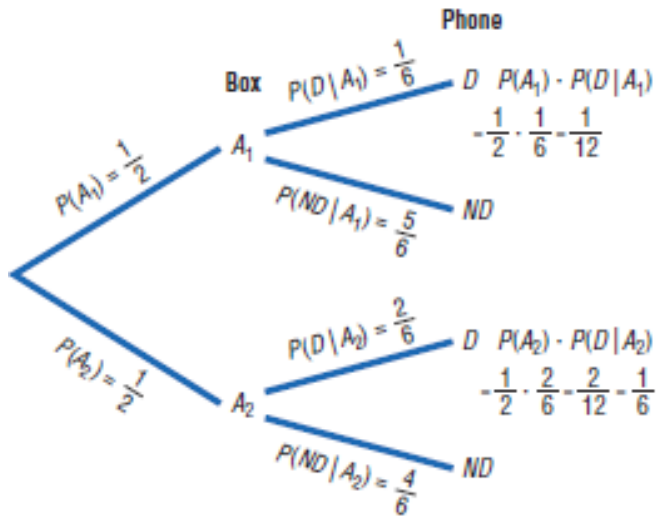
Now, a different question can be asked:

If the ball is red, what is the probability it came from box 1? In this case, the outcome is known, a red ball was selected, and you are asked to find the probability that it is a result of a previous event, that it came from box 1.

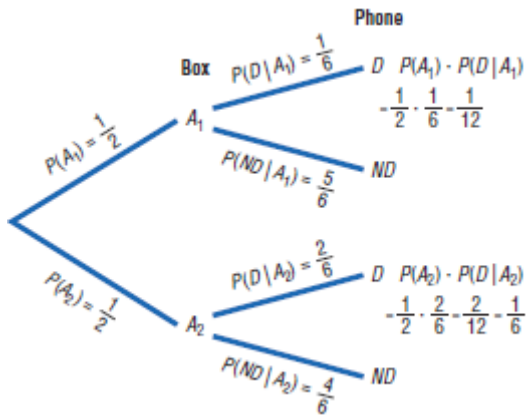
$$P(\text{Box1 given red ball}) = P(B1/R) = \frac{P(B1 \cap R)}{P(R)}$$

A shipment of two boxes, each containing six telephones, is received by a store. Box 1 contains one defective phone, and box 2 contains two defective phones. After the boxes are unpacked, a phone is selected and found to be defective. Find the probability that it came from box 2.

- 1 Select the proper notation. Let A_1 represent box 1 and A_2 represent box 2. Let D represent a defective phone and ND represent a phone that is not defective.
- 2 Draw a tree diagram and find the corresponding probabilities for each branch. The probability of selecting box 1 is $\frac{1}{2}$, and the probability of selecting box 2 is $\frac{1}{2}$. Since there is one defective phone in box 1 $\frac{1}{6}$, the probability of selecting it is $\frac{1}{6}$. The probability of selecting a nondefective phone from box 1 is $\frac{5}{6}$. Since there are two defective phones in box 2, the probability of selecting a defective phone from box 2 is $\frac{2}{6}$, or $\frac{1}{3}$; and the probability of selecting a non defective phone is $\frac{4}{6}$, or $\frac{2}{3}$. The tree diagram is



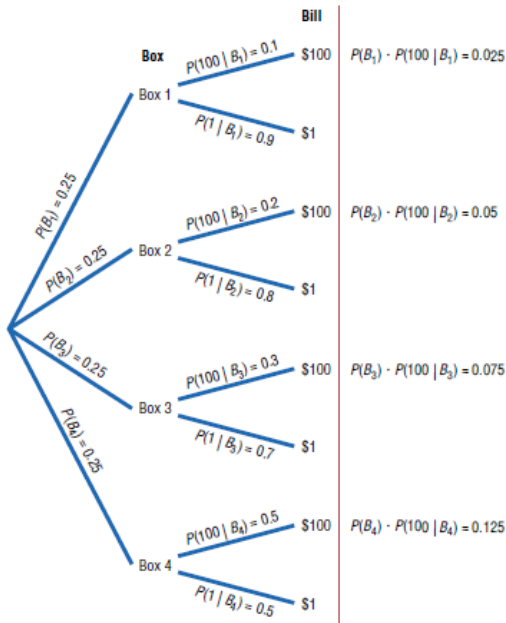
Write the corresponding formula. Since the example is asking for the probability that, given a defective phone, it came from box 2, the corresponding formula is as shown. $P(A_2/D) = ?$



$$P(A_2/D) = ?$$

$$P(A_2|D) = \frac{P(A_2) \cdot P(D|A_2)}{P(A_1) \cdot P(D|A_1) + P(A_2) \cdot P(D|A_2)}$$

On a game show, a contestant can select one of four boxes. Box 1 contains one \$100 bill and nine \$1 bills. Box 2 contains two \$100 bills and eight \$1 bills. Box 3 contains three \$100 bills and seven \$1 bills. Box 4 contains five \$100 bills and five \$1 bills. The contestant selects a box at random and selects a bill from the box at random. If a \$100 bill is selected, find the probability that it came from box 4.



$$\begin{aligned}
 P(B_4|100) &= \frac{P(B_4) \cdot P(100|B_4)}{[P(B_1) \cdot P(100|B_1) + P(B_2) \cdot P(100|B_2) \\
 &\quad + P(B_3) \cdot P(100|B_3) + P(B_4) \cdot P(100|B_4)]} \\
 &= \frac{0.125}{0.025 + 0.05 + 0.075 + 0.125} \\
 &= \frac{0.125}{0.275} = 0.455
 \end{aligned}$$

Solution

- Box 1: 1 \$100 bill, 9 \$1 bills
- Box 2: 2 \$100 bills, 8 \$1 bills
- Box 3: 3 \$100 bills, 7 \$1 bills
- Box 4: 5 \$100 bills, \$1 bills

We need to find the probability that the \$100 bill came from Box 4 given that a \$100 bill was selected. Let:

- Event A: Selecting a \$100 bill
- Event B: Selecting a bill from Box 4

We'll use Bayes' theorem:

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

First, we calculate $P(A)$, the overall probability of selecting a \$100 bill:

$$P(A_1) = \frac{1}{10}, \quad P(A_2) = \frac{2}{10}, \quad P(A_3) = \frac{3}{10}, \quad P(A_4) = \frac{5}{10}$$

$$P(A) = P(A_1) + P(A_2) + P(A_3) + P(A_4) = \frac{11}{40}$$

Calculating $P(A_1)$

- To calculate $P(A_1)$, the probability of selecting a \$100 bill from Box 1:

$$P(A_1) = \frac{\text{Number of \$100 bills in Box 1}}{\text{Total number of bills in Box 1}} = \frac{1}{10}$$

- To calculate $P(A_2)$, the probability of selecting a \$100 bill from Box 2:

$$P(A_2) = \frac{\text{Number of \$100 bills in Box 2}}{\text{Total number of bills in Box 2}} = \frac{2}{10} = \frac{1}{5}$$

- To calculate $P(A_3)$, the probability of selecting a \$100 bill from Box 3:

$$P(A_3) = \frac{\text{Number of \$100 bills in Box 3}}{\text{Total number of bills in Box 3}} = \frac{3}{10}$$

- To calculate $P(A_4)$, the probability of selecting a \$100 bill from Box 4:

$$P(A_4) = \frac{\text{Number of \$100 bills in Box 4}}{\text{Total number of bills in Box 4}} = \frac{5}{10} = \frac{1}{2}$$

Then, we calculate $P(B|A)$:

$$P(B|A) = \frac{\frac{1}{2} \times \frac{1}{4}}{\frac{11}{40}} = \frac{5}{11}$$

A real estate agent has 8 master keys to open several new homes. Only 1 master key will open any given house. If 40% of these homes are usually left unlocked, what is the probability that the real estate agent can get into a specific home if the agent selects 3 master keys at random before leaving the office?

The real estate agent can get into a specific home in following conditions

(1) $\underbrace{\text{The home is unlocked.}}_U$

(2) $\underbrace{\text{The home is locked}}_L \text{ but } \underbrace{\text{he chose an appropriate key.}}_K$

$$P \left[\begin{array}{c} \text{Estate agent} \\ \text{can get into} \\ \text{a specific} \\ \text{home} \end{array} \right] = \underbrace{P(U)}_{0.4} + \underbrace{P(L \cap K)}_{P(L) \cdot P(K)},$$

$$P(K) = \frac{(\text{selecting 1 appropriate key})(\text{Selecting 2 keys out of 7})}{\text{choosing 3 out of 8 keys}}$$

A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

- 1 What is the probability that neither is available when needed?
- 2 What is the probability that a fire engine is available when needed?

Solution:

1

$$P(A_1^c \cap A_2^c) = P(A_1^c) \cdot P(A_2^c) = \{1 - P(A_1)\} \cdot \{1 - P(A_2)\}$$

2

P(at least one fire engine is available)

A producer of a certain type of electronic component ships to suppliers in lots of twenty. Suppose that 60% of all such lots contain no defective components, 30% contain one defective component, and 10% contain two defective components. A lot is picked, two components from the lot are randomly selected and tested, and neither is defective.

- 1 What is the probability that zero defective components exist in the lot?
- 2 What is the probability that one defective exists in the lot?
- 3 What is the probability that two defectives exist in the lot?

Let D_i denote lot contain i defective components

$$P(D_0) = 0.6, \quad P(D_1) = 0.3, \quad P(D_2) = 0.1$$

$N = \{\text{A lot is picked, two components from the lot are randomly selected and tested, and neither is defective.}\}$ $P(D_0/N) = ?$, $P(D_1/N) = ?$, $P(D_2/N) = ?$

- The probability that 2 nondefective components are chosen from a lot with 0 defectives is:

$$P(N|D_0) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{components} \end{array}} = \frac{\binom{20}{2}}{\binom{20}{2}} = 1 \quad (2)$$

- The probability that 2 nondefective components are chosen from a lot with 1 defectives is:

$$P(N|D_1) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 19} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{components} \end{array}} = \frac{\binom{19}{2}}{\binom{20}{2}} = \frac{171}{190} = 0.9 \quad (3)$$

•

$$P(N|D_2) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 18} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{components} \end{array}} = \frac{\binom{18}{2}}{\binom{20}{2}} = \frac{153}{190} \approx 0.8 \quad (4)$$

$$\begin{aligned}
 P(D_0|N) &= \frac{P(N|D_0)P(D_0)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{1 \cdot 0.6}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.6}{0.95} \\
 &\approx \mathbf{0.63}
 \end{aligned}$$

$$\begin{aligned}
 P(D_1|N) &= \frac{P(N|D_1)P(D_1)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{0.9 \cdot 0.3}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.27}{0.95} \\
 &\approx \mathbf{0.284}
 \end{aligned}$$

$$\begin{aligned}
 P(D_2|N) &= \frac{P(N|D_2)P(D_2)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{0.8 \cdot 0.1}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.08}{0.95} \\
 &\approx \mathbf{0.084}
 \end{aligned}$$

A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?

- Let C be the event that the women over 60 has certain form of cancer $P(C) = 0.07$.
- Let A denotes that test results in positive.
- It is known that 10% of the time the test gives a false negative $P(A^c/C) = 0.1$
- It is known that 5% of the time the test gives a false positive $P(A/C^c) = 0.05$.

$$P(C/A^c) = ?$$

$$P(C/A^c) = \frac{P(A^c/C)P(C)}{P(A^c/C^c)P(C^c) + P(A^c/C)P(C)}$$

$$P(A^c/C^c) = 1 - P(A/C^c) = 1 - .005 = 0.95$$

$$\begin{aligned} &= \frac{0.1 \cdot 0.07}{0.95 \cdot 0.93 + 0.1 \cdot 0.07} \\ &= \frac{0.007}{0.8835 + 0.007} \\ &= \frac{0.007}{0.8905} \approx 0.0079 \end{aligned}$$

QUIZ #1 & QUIZ # 2

- ① An urn contains 5 red balls and 4 green balls. A ball is drawn.
 - (a) If it's green a two red balls are added to the urn and if it's red a one green ball and one red ball are add from the urn. (The original ball is not returned to the urn.) Then a second ball is drawn. What is the probability the second ball is red?
 - (b) If it's green a red ball is added to the urn and if it's red a green ball is added to the urn. (The original ball is not returned to the urn.) Then a second ball is drawn. What is the probability the that both balls are of different colors?
- ② For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15. What is the probability that
 - (i) at least one member of a married couple will vote?
 - (ii) a wife will vote, given that her husband will vote?
 - (iii) a husband will vote, given that his wife will not vote?

- ① A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss.
- ② The probability that Sam parks in a no-parking zone and gets a parking ticket is 0.06, and the probability that Sam cannot find a legal parking space and has to park in the no parking zone is 0.20. On Tuesday, Sam arrives at school and has to park in a no-parking zone. Find the probability that he will get a parking ticket.

Quiz# 2,3 for Odd Rows

The probability that a vehicle entering the Luray Caverns (LC) has Canadian license plates is 0.12; the probability that it is a camper is 0.28; and the probability that it is a camper with Canadian license plates is 0.09. What is the probability that

- (a) a camper entering the LC has Canadian license plates?
- (b) a Canadian license plates vehicle entering the LC is a camper?
- (c) a vehicle entering the LC does not have Canadian plates or is not a camper?

Quiz# 2,3 for Even Rows

One overnight case contains 2 bottles of aspirin and 3 bottles of thyroid tablets. A second tote bag contains 3 bottles of aspirin, 2 bottles of thyroid tablets, and 1 bottle of laxative tablets. If 1 bottle of tablets is taken at random from each piece of luggage, find the probability that

- (i) both bottles contain thyroid tablets:
- (ii) neither bottle contains thyroid tablets;
- (iii) the 2 bottles contain different tablets.

Permutation and Combination

Permutations deal with arrangements where the
order matters,

combinations deal with selections where the
order **does not** matter.

Permutations $P(n, r)$

The number of ways to **arrange** r items from n distinct items, where the order matters.

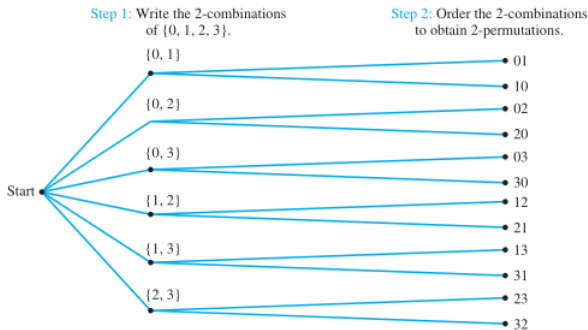
Useful for scenarios like seating arrangements, race results, or creating passwords.

Combinations $C(n, r)$

The number of ways to **choose** r items from n distinct items, where the order does not matter.

Relevant for scenarios like forming committees, lottery draws, or choosing toppings for a pizza.

Write all 2-permutations of the set $\{0, 1, 2, 3\}$?



$$P(4, 2) = C(4, 2) \cdot 2!$$

3 elements subsets for

$\{0, 1, 2, 3\}$

① $\{0, 1, 2\}$

② $\{0, 1, 3\}$

③ $\{0, 2, 3\}$

④ $\{1, 2, 3\}$

$$C_3^4 = \frac{4!}{3! \cdot 1!} = 4$$

3-Element Subsets and Permutations for $\{0, 1, 2, 3\}$

- Subset: $\{0, 1, 2\}$

- 1 0 1 2
- 2 0 2 1
- 3 1 0 2
- 4 1 2 0
- 5 2 0 1
- 6 2 1 0

- Subset: $\{0, 1, 3\}$

- 1 0 1 3
- 2 0 3 1
- 3 1 0 3
- 4 1 3 0
- 5 3 0 1
- 6 3 1 0

- Subset: $\{0, 2, 3\}$

- 1 0 2 3
- 2 0 3 2
- 3 2 0 3
- 4 2 3 0
- 5 3 0 2
- 6 3 2 0

- Subset: $\{1, 2, 3\}$

- 1 1 2 3
- 2 1 3 2
- 3 2 1 3
- 4 2 3 1
- 5 3 1 2
- 6 3 2 1

$$P_3^4 = \frac{4!}{1!} = 4 \times 3 \times 2 = 24$$

How many Possibilities are for
5 books but 3 can be arranged on showcase?

How many Possibilities are for

5 books but 3 can be arranged on showcase?

A B C	A B D	A B E	A C B	A C D	A C E
A D B	A D C	A D E	A E B	A E C	A E D
B A C	B A D	B A E	B C A	B C D	B C E
B D A	B D C	B D E	B E A	B E C	B E D
C A B	C A D	C A E	C B A	C B D	C B E
C D A	C D B	C D E	C E A	C E B	C E D
D A B	D A C	D A E	D B A	D B C	D B E
D C A	D C B	D C E	D E A	D E B	D E C
E A B	E A C	E A D	E B A	E B C	E B D
E C A	E C B	E C D	E D A	E D B	E D C

To establish the formula for permutations, consider we have n distinct objects and we want to arrange them in a specific order, selecting r objects at a time:

- ① **1st Selection:** n choices for the first object.
- ② **2nd Selection:** After selecting the 1st object, $n - 1$ choices for the 2nd object.
- ③ **3rd Selection:** We have $n - 2$ choices for the third object.
- ④ **And so on...:** This process continues until we've selected r objects.

Therefore, the total number of permutations of n objects taken r at a time is the product of the number of choices at each step:

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \dots \times (n - r + 1)$$

Simplified using factorials:

$$P(n, r) = \frac{n!}{(n - r)!}$$

Combinations

Consider the scenario where we want to select r objects out of n . We use the formula for combinations:

$$C(n, r) = \frac{n!}{r! \cdot (n - r)!}$$

Proof

Consider selecting r objects out of n without regard to the order. This is the number of subsets of size r that can be formed from a set of n objects.

- The number of subsets of size r from a set of n objects is given by the binomial coefficient, often denoted as $\binom{n}{r}$ or $C(n, r)$.
- The formula for the binomial coefficient is:

$$\binom{n}{r} = \frac{n!}{r! \cdot (n - r)!}$$

How many unordered selections of two elements can be made from the set $\{0, 1, 2, 3\}$?

$\{0, 1\}, \{0, 2\}, \{0, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$

$$C(n, r) = C(4, 2) = \frac{4!}{2!(4-2)!} = \frac{4 \times 3 \times 2!}{2! \times 2!} = 6$$

Combination

- 1 Combine r objects out of n objects
- 2 Order/position does not matter

Permutation

- 1 Combine r objects out of n objects
- 2 Arrange r objects in order

“How many ways are there to arrange 5 boys and 3 girls in a row so that no two girls are adjacent?”

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the **order of these elements matters**.

Many other counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the **order of the elements selected does not matter**.

For example, in how many ways can we select three students from a group of five students to stand in line for a picture? How many different committees of three students can be formed from a group of four students?

Permutations	Combinations
If n and r are integers with $0 \leq r \leq n$, then $P(n, r) = \frac{n!}{(n - r)!}$	The number of r-combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, equals $C(n, r) = \frac{n!}{r!(n - r)!}.$

Permutation or Combinations or what?

A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits. How many different PINs are possible?

- ① With repetition allowed
- ② Repetition not allowed

- ① How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?
- ② In how many ways can 5 differently colored marbles be arranged in a row?
- ③ Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?
- ④ How many permutations of the letters ABCDEFGH contain the string ABC ?
- ⑤ How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?
- ⑥ How many eight-bit strings have exactly three 1's?
- ⑦ How many bit strings of length n contain exactly r 1s?

A committee of 3 members is to be formed consisting of one representative each from labor, management, and the public. If there are 3 possible representatives from labor, 2 from management, and 4 from the public, determine how many different committees can be formed using (a) the fundamental principle of counting and (b) a tree diagram.

Suppose the group of twelve consists of five men and seven women.

- ① How many five-person teams can be chosen that consist of three men and two women?
- ② How many five-person teams contain at least one man?
- ③ How many five-person teams contain at most one man?

(a) To answer this question, think of forming a team as a two-step process:

Step 1: Choose the men.

Step 2: Choose the women.

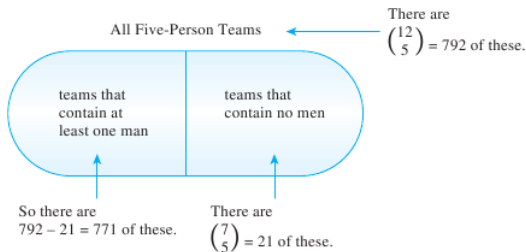
There are $\binom{5}{3}$ ways to choose the three men out of the five and $\binom{7}{2}$ ways to choose the two women out of the seven. Hence, by the product rule, $\binom{5}{3} \cdot \binom{7}{2} = 210$

How many five-person teams contain at least one man?

Observe that the set of five-person teams containing at least one man equals the set difference between the set of all five-person teams and the set of five-person teams that do not contain any men.

How many five-person teams contain at least one man?

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Alternative way

There are

$\binom{5}{1} \binom{7}{4}$ teams with one man and four women

$\binom{5}{2} \binom{7}{3}$ teams with two men and three women

$\binom{5}{3} \binom{7}{2}$ teams with three men and two women

$\binom{5}{4} \binom{7}{1}$ teams with four men and one woman

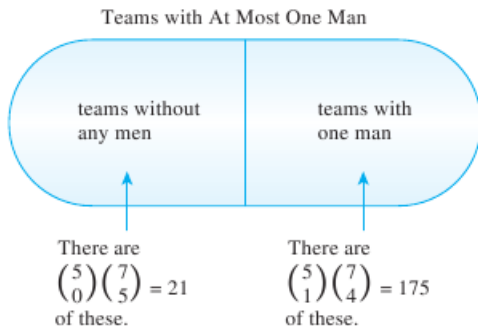
$\binom{5}{5} \binom{7}{0}$ teams with five men and no women.

Hence, by the addition rule,

$$\begin{aligned} & \left[\begin{array}{l} \text{number of teams with} \\ \text{at least one man} \end{array} \right] \\ &= \binom{5}{1} \binom{7}{4} + \binom{5}{2} \binom{7}{3} + \binom{5}{3} \binom{7}{2} + \binom{5}{4} \binom{7}{1} + \binom{5}{5} \binom{7}{0} \end{aligned}$$

How many five-person teams contain at most one man?

How many five-person teams contain at most one man?



So the total number of teams with at most one man is $21 + 175 = 196$.

Permutations with Indistinguishable Objects

How many different strings can be made by reordering the letters of the word SUCCESS?

Permutations with Indistinguishable Objects

How many different strings can be made by reordering the letters of the word SUCCESS?

Because some of the letters of SUCCESS are the same, the answer is not given by the number of permutations of seven letters. This word contains **three Ss, two Cs, one U, and one E.**

Permutations with Indistinguishable Objects

How many different strings can be made by reordering the letters of the word SUCCESS?

Because some of the letters of SUCCESS are the same, the answer is not given by the number of permutations of seven letters. This word contains **three Ss, two Cs, one U, and one E**. To determine the number of different strings that can be made by reordering the letters, **first note that the three Ss can be placed among the seven positions in $C(7, 3)$ different ways, leaving four positions free**. Then the two Cs can be placed in $C(4, 2)$ ways, leaving two free positions. **The U can be placed in $C(2, 1)$ ways, leaving just one position free**. Hence E can be placed in $C(1, 1)$ way. Consequently, from the product rule, the number of different strings that can be made is

$$C(7, 3)C(4, 2)C(2, 1)C(1, 1) = \frac{7!}{3!.4!} \cdot \frac{4!}{2!.2!} \cdot \frac{2!}{1!.1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420$$

Consider various ways of ordering the letters in the word MISSISSIPPI: IIMSSPISSIP, ISSSPMIIPIS, PIMISSSSIIIP, and so on. How many distinguishable orderings are there?

Step 1: Choose a subset of four positions for the S 's.

Step 2: Choose a subset of four positions for the I 's.

Step 3: Choose a subset of two positions for the P 's.

Step 4: Choose a subset of one position for the M .

1 2 3 4 5 6 7 8 9 10 11

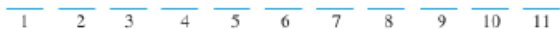
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Since there are 11 positions in all, there are $\binom{11}{4}$ subsets of four positions for the S 's. Once the four S 's are in place, there are seven positions that remain empty, so there are $\binom{7}{4}$ subsets of four positions for the I 's. After the I 's are in place, there are three positions left empty, so there are $\binom{3}{2}$ subsets of two positions for the P 's. That leaves just one position for the M . But $1 = \binom{1}{1}$. Hence by the multiplication rule,

$$[\text{No. of ways to position all letters}] = C_4^{11} C_4^7 C_2^3 C_1^1 = \frac{11!}{4!4!2!1!} = 34650.$$

Theorem

Suppose a collection consists of n objects of which

n_1 are of type 1 and are indistinguishable from each other

n_2 are of type 2 and are indistinguishable from each other

$\dots$

$\dots$

$\dots$

n_k are of type k and are indistinguishable from each other,

and suppose that $n_1 + n_2 + \dots + n_k = n$. Then the number of distinguishable permutations of the n objects is

$$\binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-n_2-\dots-n_{k-1}}{n_k} = \frac{n!}{n_1! n_2! n_3! \dots n_k!}.$$

A box contains 24 transistors, 4 of which are defective. If 4 are sold at random, find the number of ways for the following

- a. Exactly 2 are defective.
- b. None is defective.
- c. All are defective.
- d. At least 1 is defective

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(a) ${}^{24}C_2$ is wrong answer.

There are two types of transistors "Defective" and "Non-defective". We have to choose exactly 2 defective out of 4 defective transistors.

(b)

(c)

(d)

In how many ways can 10 objects be split into two groups containing 4 and 6 objects, respectively?

How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards?

To begin, note that the first player can be dealt 5 cards in $C(52, 5)$ ways. The second player can be dealt 5 cards in $C(47, 5)$ ways, because only 47 cards are left. The third player can be dealt 5 cards in $C(42, 5)$ ways. Finally, the fourth player can be dealt 5 cards in $C(37, 5)$ ways. Hence, the total number of ways to deal four players 5 cards each is

$$C(52, 5)C(47, 5)C(42, 5)C(37, 5) = \frac{52!}{5!5!5!32!}.$$

Write in Mathematical Form and Then Solve

- 1 A card is drawn at random from an ordinary deck of 52 playing cards. Find the probability that it is neither a four nor a club.
- 2 A ball is drawn at random from a box containing 6 red balls, 4 white balls, and 5 blue balls. Determine the probability that it is red or white.
- 3 Determine the probability of three 6s in 5 tosses of a fair die.
- 4 A and B play 12 games of chess of which 6 are won by A, 4 are won by B, and 2 end in a draw. They agree to play a tournament consisting of 3 games. Find the probability that (a) A and B win alternately, (b) B wins at least 1 game.